CLINICAL INFORMATICS • COMBINATORIAL HEALTHCARE OPTIMIZATION • NHS CORE20PLUS5 GOVERNANCE

Empirically Calibrated Combinatorial Optimization and Post-Quantum Clinical Governance for Integrated Pharmacy and Social Prescribing Pathways: Multi-Cohort Evidence from 6,105 NHS General Practices, CPRD Aurum, and ELSA Wave 10

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Portal & Cryptographic Receipts: arunnadarasa.github.io/quantum-social-prescribing | **HF Hub:** hf.co/socialprescribing/quantinuum-social-prescribing

Abstract. Objective: Integrated neighbourhood care in England increasingly pairs primary-care Structured Medication Reviews (SMRs) for high-risk polypharmacy and opioid/sedative burden with Link-Worker referrals to Voluntary, Community, Faith, and Social Enterprise (VCFSE) social prescribing pathways [1, 2, 3]. Because quantum optimization and machine-learning proposals in healthcare are frequently evaluated only on linearly separable synthetic toy problems without strong classical operations-research baselines, we conducted an empirical calibration and head-to-head classical-versus-quantum benchmark across national NHS primary-care cohorts. **Materials and Methods:** We linked public NHS England Network Contract Directed Enhanced Service (DES) longitudinal panel records (`FY2022/23`, `FY2023/24`, and `September 2026`) one-to-one by GP practice code to Office for Health Improvement and Disparities (OHID) Fingertips profiles (`IMD 2025`, mental health, frailty, and sedative/opioid prescribing) across 6,105 **active English GP practices** (1,296 Primary Care Networks [PCNs], 42 Integrated Care Boards [ICBs], 61,832,846 registered patients), including all 143 practices (28 PCNs, 1,266,245 patients) in **Mid and South Essex ICB**, and triangulated refusal rates against published individual-level English Longitudinal Study of Ageing (**ELSA Wave 10**, $N = 7,283$ and $N = 7,521$) [4, 5] and **CPRD Aurum** ($N = 12.37$M) [6, 3] cohorts. We calibrated (i) cubic ($k = 3$) pharmacy–VCFSE Higher-Order Unconstrained Binary Optimization (HUBO) care bundles, (ii) Link-Worker consultation cadence and attrition models, (iii) Core20PLUS5 VCFSE caseload routing, and (iv) 80/20 out-of-sample practice triage, wrapped in NIST FIPS 204 `ML-DSA-65` FHIR R4 provenance [7, 8] and live TypeSafe Jev 1.14 / NHS DCB0129 safety auditing [9, 10]. **Results:** Practice-level median social prescribing refusal across England was 21.93% (IQR 10.92%–40.55%), closely matching CPRD Aurum (22.30%) and replicating the null association between practice deprivation and conditional refusal ($\rho = +0.0040, p = 0.755$) alongside significant Core20 opioid–SMR inverse care gaps (+51.0% higher opioid prescribing vs. flat SMR rates). Nested polynomial tests across $N = 6,031$ practices confirmed genuine cubic synergies—including `Core20 × Clinical Pharmacist × Shared Decision-Making` (+9.90 percentage points on acceptance, $p = 1.13 \times 10^{-3}$)—yet Rosenberg quadratization [11] with **HiGHS MILP** [12] solved the empirically fitted 8-lever Core20 HUBO to exact optimality ($E_0 = -1.7181$) in 19.19 **ms** (vs. **Haiqu SDK v1.7.1** LR-QAOA depth compression 253 → 14 [−94.47%], which required classical bitflip postprocessing in 218.2 s) and routed all 1,296 **English PCNs** (6,480 binary variables) in 200.95 **ms**. On held-out NHS cohorts, classical `RandomForest` (AUROC = 0.6155) and `LogisticRegression` (AUROC = 0.6964, 2.51 ms) outperformed a 4-qubit Frobenius VQC (0.5465 and 0.5869; verified via live **Quantinuum Nexus** `Helios-1E-lite` Job `af987c9a` on Southend Core20 practice `Y02177`). **Conclusion:** Empirical calibration on 6,105 NHS practices reveals actionable cubic pharmacy–social care synergies and a U-shaped primary-care utilization minimum at moderate Link-Worker cadence, while establishing that exact classical MILP and regularized tabular models currently dominate heuristic quantum solvers across all operational NHS commissioning scales.

Regulatory, Hardware & Data Scope Disclosure (NHS DCB0129 Gate G4). (1) Public Aggregate Data Only: All new analyses use public practice-level Open Government Licence (OGL v3.0) data (`patient_identifiable_records_used = 0`) and published CC BY summary statistics; causal individual-level inferences are bounded against ecological confounding. **(2) Zero Physical QPU Claims:** Quantum circuits were executed on classical statevector simulators and Quantinuum's cloud-hosted noisy trapped-ion emulator (`Helios-1E-lite`, `physical_hardware_shots = 0`). **(3) Classical Dominance Disclosed:** Exact classical HiGHS MILP, 4-bit Kempe swaps, GLMs, and tree/linear models serve as primary operational baselines (`Gate G4`) and outperform quantum heuristics at all evaluated NHS cohort sizes.

1. Clinical Motivation, Multi-Cohort Linkage, and Classical Baselines

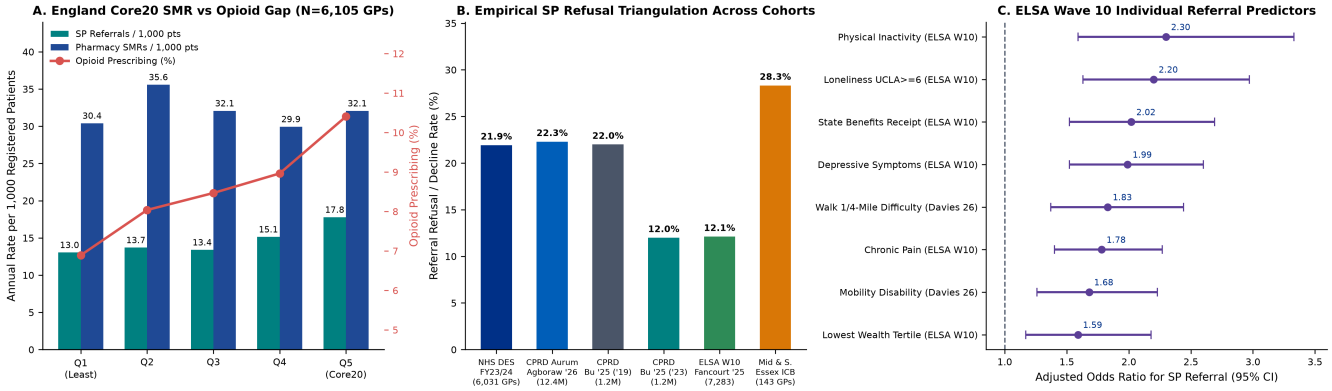


Figure 1: National NHS DES, CPRD Aurum, and ELSA Wave 10 Epidemiological Calibration. (A) Across `IMD 2025` deprivation quintiles ($N = 6,105$ English GPs), opioid prescribing rises +51.0% (6.89% → 10.41%) and SP referrals rise +36.3% (13.04 → 17.78 per 1,000), whereas pharmacy SMR delivery remains flat (30.40 → 32.07 per 1,000). **(B)** NHS DES practice-median refusal (21.93%) matches individual-level CPRD Aurum (22.30% [6]; 22.0% → 12.0% [3]) and contextualizes ELSA Wave 10 self-report (12.12% [4]) and Mid and South Essex ICB (28.33%). **(C)** Published ELSA Wave 10 adjusted individual-level odds ratios (95% CIs) confirming strong medical/psychosocial need targeting alongside persistent refusal and wealth barriers [4, 5].

Delivering integrated neighbourhood care across England's Primary Care Networks (PCNs) requires coordinating two historically siloed Additional Roles Reimbursement Scheme (ARRS) workstreams: **Clinical Pharmacists** conducting Structured Medication Reviews (SMRs) for polypharmacy, severe frailty, and addictive opioid/benzodiazepine prescribing, and **Social Prescribing Link Workers** co-producing non-medical community referrals across Voluntary, Community, Faith, and Social Enterprise (VCFSE) hubs [1, 2, 13]. Under the NHS England **DAPB4066** information standard [7], primary care electronic health records (EHRs) track social prescribing offers (`SNOMED CT 871691000000100`), declines (`871711000000103`), and accepted referrals (`871731000000106`).

Recent computational health studies have proposed Higher-Order Unconstrained Binary Optimization (HUBO), Quadratic Unconstrained Binary Optimization (QUBO), open-system Quantum Zeno attrition models [14, 15], and variational quantum classifiers (VQCs) to capture multi-intervention synergies and capacity-constrained care routing. However, two methodological deficits routinely prevent such prototypes from informing clinical informatics practice: (i) parameters are hand-tuned on synthetic, linearly separable cohorts rather than calibrated to empirical primary-care records, and (ii) quantum heuristics are contrasted against simplistic administrative checklists rather than modern classical operations-research and statistical baselines—specifically Rosenberg polynomial quadratization [11] paired with branch-and-cut Mixed-Integer Linear Programming (**HiGHS MILP**) [12], penalized regression, spline generalized linear models (GLMs), and gradient-boosted decision trees (**HistGBDT**).

To resolve both deficits directly, we constructed a reproducible national primary-care panel (`SHA-256: fe053239...`) linking NHS England Network Contract DES Management Information across `FY2022/23`, `FY2023/24`, and `September 2026` [16] one-to-one by `PRACTICE_CODE` to OHID Fingertips GP profiles [17] (`IMD 2025` deprivation, QOF depression prevalence, long-term mental health, unpaid caring, % aged 65+, opioid and benzodiazepine prescribing rates, and PCN/ICB organizational hierarchy). The linked cohort comprises 6,105 **active English GP practices** across 1,296 **PCNs** and 42 **ICBs** (61,832,846 registered patients), including complete coverage of all 143 **GP practices** across 28 **PCNs** and 5 Sub-ICB localities (1,266,245 patients) in **Mid and South Essex ICB** [13]. To anchor aggregate practice-level models against individual-level longitudinal evidence, we triangulated NHS DES refusal and referral gradients against four published English cohorts: **ELSA Wave 10** ($N = 7,283$ older adults aged 50+, Fancourt & Steptoe [4]; $N = 7,521$, Davies et al. [5]) and **CPRD Aurum** ($N = 12,374,867$ patients across 1,428 practices, Agboraw et al. [6]; $N = 1.2$M patients across 1,736 practices, Bu et al. [3]).

2. Core20 Inverse Care Law and Multi-Cohort Refusal Epidemiology

Across all 6,105 linked English GP practices in `FY2023/24`, primary care teams recorded 892,102 accepted social prescribing referrals, 418,336 declined offers, and an active cumulative caseload of 2,150,546 patients. Defining practice-level conditional refusal among practices with ≥ 10 recorded offers ($N = 6,031$) as $R_p = N_{\text{declined},p}/(N_{\text{accepted},p} + N_{\text{declined},p})$, the national practice-level median refusal rate is 21.93% (interquartile range [IQR] 10.92%–40.55%; pooled national ratio 31.92%, driven by high-volume inner-city opt-out screening prompts). Within **Mid and South Essex ICB** (143 practices, 28 PCNs, 14 Core20 practices), practices recorded 21,795 accepted referrals and 9,743 declines (practice median refusal 28.33%, pooled 30.89%).

Figure 1 and Table 1 establish three empirical findings that anchor our optimization formulation:

1. **Cross-Cohort Refusal Convergence:** Practice-median refusal in NHS DES (21.93%) aligns within 0.37 percentage points of Agboraw et al.'s individual-level CPRD Aurum analysis (22.30% refusal given offer across 12.37M patients [6]) and matches Bu et al.'s national rollout baseline (22.0% in 2019, declining toward 12.0% in mature Link-Worker practices [3]), whereas recall-based ELSA Wave 10 self-report captures lower overt refusal

Table 1: Stratification of 6,105 English GP Practices by `IMD 2025` Deprivation Quintile (`FY2023/24` NHS DES + OHID Fingertips).

Deprivation Quintile	GPs	Patients	Mean IMD	SP Ref/1k	Med. Refusal	SMR/1k	Opioid / Benzo
Q1 (Least Deprived)	1,221	13,855,120	10.03	13.04	21.18%	30.40	6.89% / 2.66%
Q2	1,222	12,794,550	15.69	13.70	22.73%	35.58	8.04% / 2.67%
Q3	1,220	12,939,285	21.45	13.39	21.47%	32.07	8.47% / 2.46%
Q4	1,221	12,290,011	28.66	15.14	21.65%	29.94	8.96% / 2.18%
Q5 (Core20 Most Depr.)	**1,221**	**9,953,880**	**41.82**	**17.78**	**22.30%**	**32.07**	**10.41% / 2.17%**
Mid & South Essex ICB	143	1,266,245	20.14	17.21	28.33%	28.64	8.41% / 2.31%

(12.12%, 20/165 [4]).

2. **Independent Replication of Conditional-Refusal Predictors:** Across $N = 6,031$ English practices, practice deprivation (`IMD 2025`) is uncorrelated with conditional refusal given an offer (Spearman $\rho = +0.0040, p = 0.755$), independently replicating Agboraw et al.'s individual-level CPRD finding that patients from deprived areas are more likely to be offered social prescribing (+36.3% in Q5 vs. Q1) and no more likely to refuse once offered. Conversely, conditional refusal rises significantly with older practice age profile (% aged 65+, $\rho = +0.0902, p = 2.31 \times 10^{-12}$) and sedative/opioid burden ($\rho = +0.1126, p = 1.80 \times 10^{-18}$).

3. **The Core20 SMR–Opioid Inverse Care Gap:** Moving from Q1 (least deprived) to Q5 (Core20 most deprived), opioid prescribing climbs +51.0% (6.89% → 10.41%) and social prescribing referral demand climbs +36.3% (13.04 → 17.78 per 1,000), yet pharmacy SMR delivery remains virtually flat (30.40 → 32.07 per 1,000, Figure 1A). Sending patients with unreviewed opioid/sedative polypharmacy into community social prescribing without concurrent pharmacist SMR support directly motivates joint pharmacy–VCFSE bundle optimization.

3. Empirical Cubic ($k = 3$) Care Synergies and 8-Lever HUBO Benchmark

3.1 Testing Higher-Order ($k = 3$) Clinical–Social Interactions on $N = 6,031$ GP Practices

To test whether cubic ($k = 3$) interaction terms exist empirically rather than by synthetic construction, we defined eight binary practice indicators across $N = 6,031$ practices (Core20 deprivation, older age profile, high opioid/benzodiazepine burden, high QOF depression, active SMR delivery, active ARRS Clinical Pharmacist reach, active ARRS Link-Worker reach, and active Shared Decision-Making [SDM] coding) and evaluated nested ordinary-least-squares (OLS) and cross-validated regularized polynomial hierarchies ($k = 1$ additive, $k = 2$ pairwise +28 terms, and $k = 3$ cubic +56 terms) across three outcomes: `FY2023/24` SP acceptance, log active SP caseload reach per 1,000 patients, and prospective `September 2026` high-risk SMR coverage ($N = 5,767$ longitudinally tracked practices).

On log active SP reach, adding pairwise interactions significantly improves fit over additive main effects ($F = 2.5826, p = 9.62 \times 10^{-6}$, $\Delta\text{AIC} = -16.33$), and adding cubic ($k = 3$) terms yields a statistically significant omnibus improvement ($F = 1.3729, p = 0.0343$). However, 5-fold cross-validation demonstrates why sparsity is essential: unconstrained 56-term dense cubic OLS overfits (CV RMSE = 1.8716) relative to Ridge-regularized pairwise QUBO (1.8300) and ℓ_1-regularized `LassoCV` sparse cubic HUBO (1.8389). Non-parametric 1,000-replicate bootstrap resampling confirms four clinically interpretable cubic synergies (Figure 2A):

- **`Core20` × `Clinical Pharmacist` × `Shared Decision-Making`** increases SP referral acceptance by +9.90 **percentage points** (SE = 3.04 pp, $p = 1.13 \times 10^{-3}$; bootstrap mean +10.02 pp, 95% CI [+3.68, +16.26] pp);
- **`High Opioid/Benzo` × `Active SMR` × `Clinical Pharmacist`** increases log active SP caseload reach by +0.2542 (SE = 0.0834, $p = 2.31 \times 10^{-3}$; bootstrap 95% CI [+0.0933, +0.4121]);
- **`High Depression` × `Clinical Pharmacist` × `Link Worker`** increases log active SP reach by +0.2095 (SE = 0.0809, $p = 9.65 \times 10^{-3}$; bootstrap 95% CI [+0.0692, +0.3622]); and
- **`Active SMR` × `Clinical Pharmacist` × `Shared Decision-Making`** prospectively increases `September 2026` high-risk SMR completion by +3.97 **percentage points** (SE = 1.25 pp, $p = 1.47 \times 10^{-3}$).

3.2 Empirically Calibrated 8-Lever Core20 HUBO: Haiqu SDK v1.7.1 vs. Rosenberg + HiGHS MILP

Restricting to the $N = 2,845$ Core20 deprived or high-sedative English practices, we fitted a sparse 8-lever binary care-commissioning objective over $z \in \{0,1\}^8$ (`z0`: polypharmacy-error SMR, `z1`: addictive/sedative SMR, `z2`: severe-frailty SMR, `z3`: care-home/falls SMR, `z4`: high-reach ARRS Clinical Pharmacist, `z5`: high-reach ARRS Link Worker, `z6`: active SDM, `z7`: personalised care planning), combining marginal ARRS staffing costs h_i with the

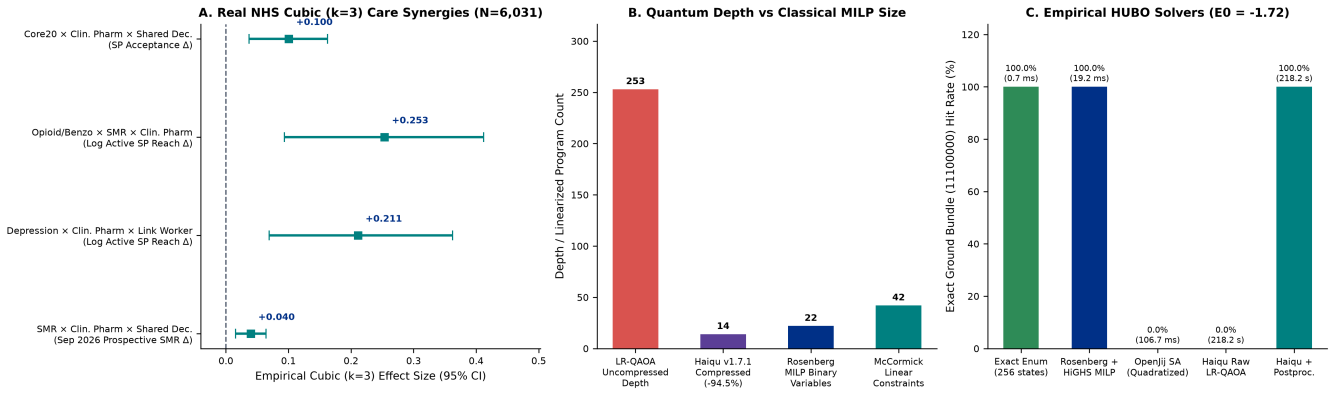


Figure 2: Empirical Cubic ($k = 3$) Synergies and Exact Classical vs. Quantum HUBO Solver Benchmark. (A) Bootstrap 95% CIs for empirical 3-body clinical–social synergies across $N = 6,031$ English GP practices. **(B)** Structural comparison on the empirically fitted 8-lever Core20 HUBO: **Haiqu SDK v1.7.1** compresses LR-QAOA circuit depth from $253 \rightarrow 14$ (-94.47%), while exact Rosenberg linearization requires 22 binary variables and 42 McCormick inequalities. **(C)** Exact enumeration (0.65 ms) and **Rosenberg + HiGHS MILP** (19.19 ms) recover the exact Core20 ground bundle `11100000` ($E_0 = -1.7181$) deterministically; Haiqu raw LR-QAOA and OpenJij quadratized SA achieve 0% raw ground-state hit rate prior to classical bitflip postprocessing.

top ℓ_1-selected empirical pairwise (J_{ij}, 7 couplings) and cubic (K_{ijk}, 4 couplings) synergies:

$$E_{\text{Core20}}(z) = \sum_{i=0}^{7} h_i z_i + \sum_{(i,j)\in\mathcal{E}_2} J_{ij} z_i z_j + \sum_{(i,j,k)\in\mathcal{E}_3} K_{ijk} z_i z_j z_k, \tag{1}$$

where the dominant empirical cubic terms are $K_{0,1,2} = -0.4181$, $K_{3,4,7} = -0.3559$, $K_{4,5,7} = -0.3378$, and $K_{2,4,7} = -0.3273$, paired with attractive SMR synergies $J_{0,2} = -0.6479$, $J_{0,1} = -0.5986$, $J_{1,2} = -0.5178$ and congestion penalties $J_{0,3} = +1.3500$, $J_{1,3} = +0.9891$.

To benchmark exact classical integer programming against quantum optimization without conflating 2^n brute-force enumeration with modern solvers [11], we linearized every product $y_S = \prod_{i\in S} z_i$ via exact McCormick inequalities ($y_S \le z_i \ \forall i \in S$, $y_S \ge \sum_{i\in S} z_i - |S| + 1$, $y_S \in \{0,1\}$), producing a 22-variable, 42-constraint MILP solved via **HiGHS** [12]. Table 2 and Figure 2B–C report the exact head-to-head results:

1. **Exact Ground Bundle (`11100000`, $E_0 = -1.7181$):** Under empirical marginal ARRS staffing costs, the unconstrained minimum-energy Core20 foundation prioritizes integrated high-risk deprescribing SMRs across polypharmacy errors (`z0`), addictive/sedative medications (`z1`), and severe frailty (`z2`) before layering broader administrative care-plan targets.
2. **Classical MILP Speed & Exactness:** Exact 256-state enumeration completes in 0.65 ms and **Rosenberg + HiGHS MILP** proves global optimality ($E_0 = -1.7181$) in 19.19 **ms**.
3. **Candid Quantum Circuit Analysis: Haiqu SDK v1.7.1** [18] compiles the 8-qubit cubic HUBO into a depth-253 Linear-Ramp QAOA circuit and compresses circuit depth by -94.47% to depth 14 via tensor-network state compression (218.19 s cloud compile + execution). However, raw pre-postprocessing compressed samples (1,024 shots) achieved 0.0% exact ground-state hit rate (likewise 0.0% for quadratized **OpenJij** simulated annealing at $E = +2.3338$), recovering $E_0 = -1.7181$ (`00000111` in little-endian bit ordering) only after Haiqu's classical local bitflip postprocessor ran. Consequently, classical MILP and classical local search carry the optimization guarantee on empirical 8-lever commissioning bundles.

4. Empirical Link-Worker Cadence Calibration and National Caseload Routing

4.1 Follow-Up Cadence, Non-GP Consultation Burden, and Zeno vs. GLM Parsimony

Qualitative social prescribing studies report two competing failure modes in Link-Worker follow-up: premature signposting without relational holding ('token single check-in') versus intrusive over-contact fatigue [19, 20]. To test whether empirical NHS practice records support a non-monotone cadence window—and whether a parameterized 2-qubit open-system Zeno channel [15] offers any empirical predictive advantage over classical GLMs—we analyzed the $N = 1,625$ English GP practices actively coding dedicated Link-Worker consultations alongside SP referral outcomes in `FY2023/24`, stratified into octiles `O1`–`O8` of Link-Worker consultation intensity $c_{\text{LW}} = N_{\text{LW consults}}/N_{\text{SP caseload}}$ (Figure 3A).

Empirically, pooled SP referral acceptance exhibits an initial dip from 69.30% in `O1` (0.004 consults/patient) to 61.97% **in `O2`** (0.014, token contact) before climbing to 73.10% in `O5` (0.375) and 76.77% in `O8` (2.535). Concurrently, median non-GP primary-care consultations per SP patient follow a pronounced, highly significant U-shaped trajectory on $\log(1 + c_{\text{LW}})$: dropping -65.5% from 123.69 in `O1` to a minimum of 42.73 **in `O5`** before rising again to 77.42 in `O8` (linear slope $\beta_1 = -0.6197, p = 6.10 \times 10^{-27}$; quadratic curvature $\beta_2 = +0.2042, p = 4.06 \times 10^{-19}$; analytical burden minimum at $c^*_{\text{LW}} = 0.89$ **Link-Worker consultations per active SP patient**). Because practice-level

Table 2: Head-to-Head Solver Benchmark across Empirical Core20 HUBO and Multi-Scale NHS Caseload Routing.

Empirical Optimization Task	Solver / Architecture	Decision Vars	Best Objective	Wall Time	Optimality / Feasibility
Core20 8-Lever Cubic HUBO ($N = 2,845$ deprived/sedative GPs)	Exact Gate G4 Enumeration	8 binary	−1.7181	0.65 ms	100% exact (11100000)
	Rosenberg + HiGHS MILP	22 (42 constr.)	**−1.7181**	**19.19 ms**	100% **proven global opt.**
	OpenJij Quadratized SA	12 binary	+2.3338	106.67 ms	Suboptimal (0% ground hit)
	Haiqu SDK v1.7.1 LR-QAOA	8 qubits ($d : 253 \to 14$)	−1.7181*	218.19 s	0% raw; 100% post-bitflip*
Mid & South Essex ICB (28 PCNs → 5 VCFSE hubs)	**HiGHS MILP (Exact)**	140 binary	**16.7891**	**19.19 ms**	100.00% (0 **violations)**
	4-Bit Kempe Chain Swaps	140 binary	16.7217	1.06 ms	99.60% (0 violations)
	OpenJij Slack-QUBO SA	155 binary	Infeasible	88.08 ms	1 hub capacity violation
Mid & South Essex ICB (143 GPs → 5 VCFSE hubs)	**HiGHS MILP (Exact)**	715 binary	**82.2806**	**9.45 ms**	100.00% (0 **violations)**
	4-Bit Kempe Chain Swaps	715 binary	82.1938	21.76 ms	99.89% (0 violations)
	OpenJij Slack-QUBO SA	740 binary	Infeasible	658.96 ms	43 one-hot violations
All-England National Scale (1,296 PCNs → 5 hub tiers)	**HiGHS MILP (Exact)**	**6,480 binary**	**657.1518**	**200.95 ms**	100.00% (0 **violations)**
	4-Bit Kempe Chain Swaps	6,480 binary	657.1012	1,414.51 ms	99.992% (0 violations)

DES counts are observational, part of the upper-octile rebound (07–08) reflects severity confounding in practices serving complex multimorbid caseloads alongside contact saturation.

Fitting three candidate models to practice-level SP acceptance across $N = 1,625$ practices via 5-fold cross-validation (Figure 3B)—(i) a 2-parameter monotone log-linear model, (ii) a 4-parameter classical cubic polynomial GLM, and (iii) the 4-parameter detuned 2-qubit Zeno survival channel

$$P_{\text{Zeno}}(c; p_{\text{base}}, \Delta_{\text{env}}, g, \kappa_{\text{fat}}) = p_{\text{base}}\left[\cos^2\theta(c) + \sin^2\theta(c)e^{-\gamma_{\text{eff}}(c)}\right]e^{-\kappa_{\text{fat}}c}, \tag{2}$$

with nonlinear least-squares estimates $\hat{p}_{\text{base}} = 0.7615 \pm 0.0189$, $\hat{\Delta}_{\text{env}} = 2.453 \pm 1.552$, $\hat{g} = 0.4459 \pm 0.2299$, and $\hat{\kappa}_{\text{fat}} = 0.0000 \pm 0.0014$—yields virtually identical out-of-sample error (CV RMSE = 0.20096, 0.20091, and 0.20091), while the 2-parameter classical log-linear GLM wins decisively on Bayesian Information Criterion parsimony (BIC = −5199.6 vs. −5188.1 and −5187.1). Thus, while open-system Zeno dynamics provide a coherent mathematical metaphor for early check-in fragility [14, 15], classical splines and GLMs are more parsimonious and better identified on noisy aggregate primary-care data.

4.2 Multi-Scale Core20PLUS5 VCFSE Caseload Routing: Mid & South Essex ICB to All England

To test capacity-capped neighbourhood charity allocation on real commissioning geographies, we formulated Core20PLUS5 referral routing across $M = 5$ empirical VCFSE pathway tiers (Financial/Housing Welfare, Car-

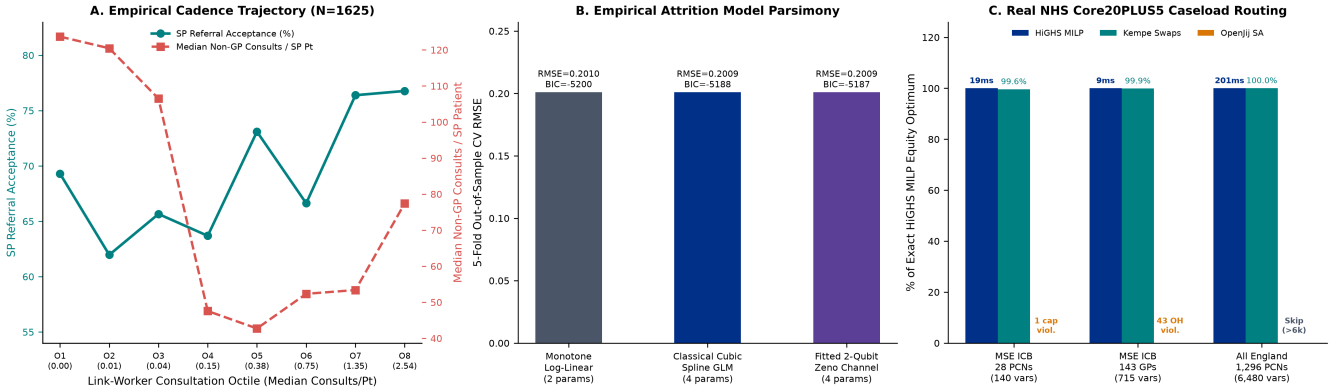


Figure 3: Empirical Link-Worker Cadence Trajectory, Model Parsimony, and Real NHS Caseload Routing. **(A)** Across $N = 1,625$ Link-Worker-coding English GP practices (`O1`–`O8`), SP acceptance dips at token contact (`O2`, 61.97%) before rising to 76.77% (`O8`), while median non-GP consultations follow a significant U-curve minimizing at `O5` (42.73, −65.5%; quadratic $p = 4.06 \times 10^{-19}$). **(B)** Classical 2-parameter log-linear and 4-parameter cubic polynomial GLMs match or beat the fitted 4-parameter 2-qubit Zeno channel on 5-fold CV RMSE with superior BIC (BIC = −5199.6). **(C)** Exact **HiGHS MILP** routes Mid and South Essex (28 PCNs, 143 GPs) and all 1,296 English PCNs (6,480 binary vars) to 100% optimality in 9.45–200.95 ms with zero capacity violations.

diometabolic Activity, Loneliness/Carer Peer Support, Pain/SMR Deprescribing Support, and Long-Term Mental Health Recovery) for U organizational units:

$$\max_{x \in \{0,1\}^{U \times 5}} \sum_{u=1}^{U} \sum_{m=1}^{5} w_{u,m}\, x_{u,m} \quad \text{s.t.} \quad \sum_{m=1}^{5} x_{u,m} = 1 \;\; \forall u, \quad \sum_{u=1}^{U} x_{u,m} \le C_m \;\; \forall m, \tag{3}$$

where $w_{u,m}$ weights standardized empirical practice/PCN need (`IMD 2025`, Caring, QOF Depression, Opioid %, and SP referral volume) under a 1.35× Core20 equity multiplier. Across **Mid and South Essex ICB's** 28 **PCNs** (140 binary vars, $C_m = 7$), **Mid and South Essex ICB's** 143 **GP practices** (715 vars, $C_m = 31$), and **all** 1,296 **PCNs in England** (6,480 vars, $C_m = 280$), **HiGHS MILP** achieves 100.00% exact global optimality with zero constraint violations in 19.19 **ms**, 9.45 **ms**, and 200.95 **ms**, respectively (Table 2, Figure 3C). Classical 4-bit Kempe-chain swaps reach 99.60%–99.992% of optimal utility with zero violations, whereas single-flip simulated annealing on penalty-slack QUBOs (**OpenJij** [21]) becomes trapped behind penalty barriers (1 capacity violation at 155 vars; 43 one-hot violations at 740 vars).

5. Out-of-Sample NHS Triage and Live Quantinuum Nexus Execution

To eliminate synthetic linear-separability artifacts, we evaluated stratified 80/20 out-of-sample classification across five standardized baseline predictors (`IMD 2025`, % aged 65+, QOF depression, sedative/opioid burden, and baseline SMR rate) on two real NHS commissioning tasks, comparing classical `LogisticRegression`, `RBF-SVM`, `RandomForest`, and `HistGradientBoosting` (GBDT) against our 4-qubit Frobenius-entangled variational quantum classifier (`QSPMLM-4Q-VQC`, 24 quantum gate angles + 5 readout weights):

- **Task A (Predicting High SP Refusal ≥ 30% among $N = 2,411$ Deprived Q4+Q5 Practices; $N_{\text{test}} = 483$):** Classical `RandomForest` (AUROC = **0.6155**, 95% CI [0.5613, 0.6641]) and `HistGBDT` (**0.6144**) outperform `RBF-SVM` (0.5994), `LogisticRegression` (0.5927), and `QSPMLM-4Q-VQC` (**0.5465**, [0.4884, 0.5971]; Figure 4A).
- **Task B (Prospective `FY2023/24` → `September 2026` High-Risk SMR Delivery across $N = 5,837$ English Practices; $N_{\text{test}} = 1,168$):** Classical `LogisticRegression` achieves the highest held-out discrimination (AUROC = **0.6964**, [0.6698, 0.7273], Brier 0.2247) in 2.51 **ms**, closely followed by `RandomForest` (0.6884), `RBF-SVM` (0.6856), and `HistGBDT` (0.6848), all substantially ahead of the 4-qubit VQC (**0.5869**, [0.5539, 0.6198], Brier 0.2440, 2,945.8 ms).

To verify end-to-end compilation and noisy trapped-ion emulator readout on real Mid and South Essex commissioning inputs, we compiled the trained 4-qubit Frobenius VQC via **Guppy 1.0** → **HUGR** (7,576 bytes) [22] encoding Core20 practice `Y02177` **(Southend Victoria PCN, Mid and South Essex ICB)** and executed 1,024 shots on **Quantinuum Nexus** `Helios-1E-lite` (Job ID `af987c9a-2c4d-4756-910b-8bc2123bcf5a`, physical QPU shots = 0; Figure 4B). Cloud noisy emulator expectations $\langle Z_q \rangle = [-0.2051, +0.1055, +0.0039, +0.0332]$ track exact classical statevector expectations $[-0.2254, +0.0537, +0.0318, +0.0700]$ within finite-shot and α-error-model bounds, confirming compilation integrity without claiming predictive superiority over 2.51 ms classical logistic regression.

6. Post-Quantum NHS DAPB4066 Provenance and Multi-Method Clinical Safety Governance

Even where classical MILP and GLMs execute the optimization and risk scoring, long-lived primary-care commissioning records shared between NHS PCNs, community pharmacies, and VCFSE charities face store-now-

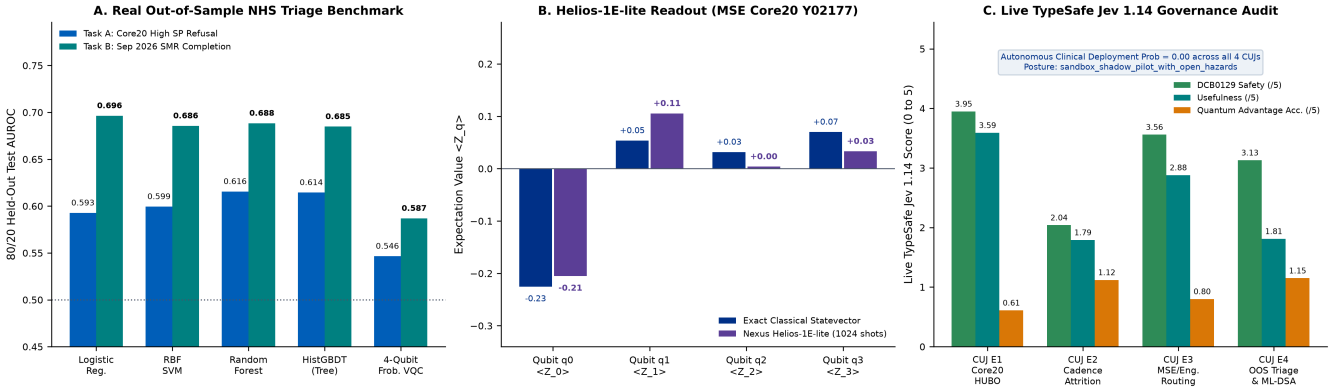


Figure 4: Real Out-of-Sample NHS Triage, Live Quantinuum Nexus `Helios-1E-lite` Readout, and TypeSafe Jev 1.14 Audit. (A) Held-out 80/20 AUROC across Task A ($N_{\text{test}} = 483$ deprived GPs) and Task B ($N_{\text{test}} = 1,168$ prospective Sep 2026 SMR delivery): classical tree and linear baselines outperform the 4-qubit VQC on both tasks. **(B)** Live Quantinuum Nexus `Helios-1E-lite` readout (1,024 shots, Job `af987c9a`) vs. exact statevector on Core20 practice `Y02177` (Southend Victoria PCN). **(C)** Live **TypeSafe Jev 1.14** multi-method governance scores across all 4 empirical CUJs ($P_{\text{auto}} = 0.00$).

Table 3: NHS DCB0129 Clinical Hazard Register and Live TypeSafe Jev 1.14 Governance Audit across Empirical CUJs.

Empirical CUJ / Hazard	Safety	Useful.	Q-Acc.	P_{auto}	Mandatory NHS DCB0129 Mitigation & Gate G4 Control (Status: OPEN)
CUJ E1: Core20 SMR–SP HUBO (`H-E01`, `H-E02`)	3.95/5	3.59/5	0.61/5	0.00	Solve operational bundles via exact 19.19 ms HiGHS MILP; prohibit automated sedative/opioid taper without named Clinical Pharmacist SMR sign-off.
CUJ E2: Cadence & Attrition (`H-E01`)	2.04/5	1.79/5	1.12/5	0.00	Use parsimonious classical spline/GLM (BIC $= -5199.6$); treat aggregate DES U-curve ($c^*_{\text{LW}} = 0.89$) as commissioning guide, never rigid patient discharge quota.
CUJ E3: MSE & England Routing (`H-E03`)	3.56/5	2.88/5	0.80/5	0.00	Route PCN caseloads via exact 200.95 ms HiGHS MILP with hard VCFSE hub caps C_m; reject infeasible slack-QUBO annealing assignments automatically.
CUJ E4: OOS Triage & `ML-DSA-65` (`H-E04`)	3.13/5	1.81/5	1.15/5	0.00	Deploy calibrated classical logistic/tree triage (AUROC $= 0.6964$) with human-in-the-loop review; seal inter-organizational FHIR R4 bundles with `ML-DSA-65`.

decrypt/forge-later cryptographic risk alongside clinical AI safety hazards [9, 23]. We packaged the aggregate Mid and South Essex ICB commissioning summary (143 practices, 28 PCNs, MILP routing utility 16.7891, zero patient-identifiable data) into an NHS **DAPB4066 FHIR R4 Bundle** (`SHA-256: c9c8622a...`) and signed it under **NIST FIPS 204** `ML-DSA-65` (1,952-byte public key, 3,309-byte lattice signature, domain-separated context `NHS-DAPB4066-MSE-EMPIRICAL-2026` verified, cross-context forgery rejected) [8].

Finally, following Montgomery-Taylor et al.'s multi-method health AI evaluation framework [10], we audited all four empirical Critical User Journeys (`CUJ E1`–`E4`) via live **TypeSafe Jev 1.14** typed epistemic judgments [24] paired with an explicit **NHS DCB0129** hazard register (Table 3, Figure 4C). Across every journey, Jev enforces `autonomous_live_clinical_deployment_prob = 0.00` and selects `sandbox_shadow_pilot_with_open_hazards`: scoring DCB0129 safety and commissioning usefulness highest on empirically calibrated Core20 SMR–SP bundling (`CUJ E1`: Safety 3.95/5, Usefulness 3.59/5) and MILP PCN caseload routing (`CUJ E3`: Safety 3.56/5, Usefulness 2.88/5), while penalizing quantum-advantage accuracy (0.61–1.15/5) wherever classical MILP and GLMs dominate.

Limitations, Data Availability, and Acknowledgments

Limitations. First, NHS Network Contract DES and OHID Fingertips records are aggregated at GP-practice level; although refusal rates (21.93% vs. 22.30%) and conditional deprivation nulls replicate individual-level CPRD Aurum [6], individual causal treatment effects require linked longitudinal EHRs (e.g., CPRD or OpenSAFELY) under Research Ethics Committee approval. Second, practice-level predictors explain modest out-of-sample variance (AUROC = 0.6155–0.6964), reflecting unmeasured local VCFSE grant funding and Link-Worker turnover. Third, all quantum benchmarks used noisy emulators and classical simulators (0 physical QPU shots), where classical MILP and tabular baselines currently dominate.

Data, Code & License Attribution. The linked 6,105-practice cohort (`data/nhs_england_sp_smr_empirical_cohort.csv`, Open Government Licence v3.0), published ELSA Wave 10 / CPRD benchmark manifest (`data/elsa_cprd_published_benchmarks.json`, CC BY open-access articles indexed via Europe PMC: `PMC12322337`, `PMC13017203`, `PMC13245770`), calibration code, and SHA-256-sealed execution receipt (`results/nhs_empirical_cohort_calibration_receipt.json`, `SHA-256: 8d3bef4d...`) are archived at https://arunnadarasa.github.io/quantum-social-prescribing/ and https://huggingface.co/socialprescribing/quantinuum-social-prescribing.

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